DEVSECOPS LAB BOOK · PHASE 5: ADVANCED TOPICS

Chapter 20

High Availability

Clustering, Replication & Failover

3	0	< 5s	4–5 hrs
HA Node Clusters	Target Data Loss (RPO)	Target Recovery (RTO)	Time Estimate

CHAPTER GOAL

Vault, Clustered

Vault runs in HA mode across multiple nodes — no single point of failure for secrets.

Database, Replicated

PostgreSQL master-slave replication keeps a live standby copy of every write.

Load Balancers, Redundant

Two or more HAProxy nodes share a virtual IP via Keepalived.

Automatic Failover

Failures are detected and traffic rerouted — no manual intervention required.

Services survive node failures. Automatic failover works. Zero-downtime deployments.

The Three Pillars

PILLAR 1



Redundancy

Multiple instances of every critical service. No single point of failure.

PILLAR 2



Failover

Automatic detection of failure and automatic redirection to a healthy instance — no manual step.

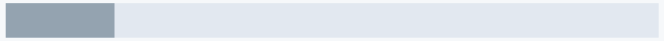
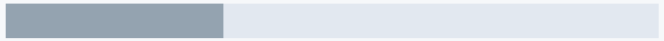
PILLAR 3



Data Replication

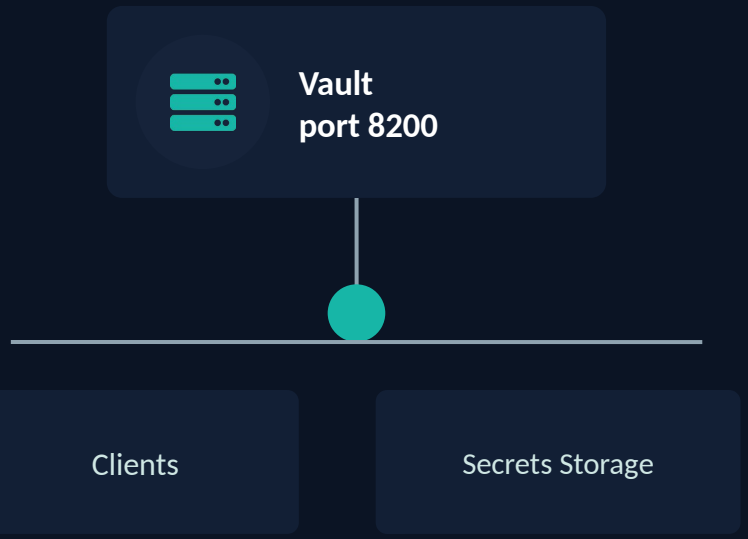
Data is synchronized across nodes so no data is lost and state stays consistent.

20.1 HIGH AVAILABILITY PRINCIPLES

UPTIME TARGET	DOWNTIME / YEAR	TIER	
99.0%	3.65 days / yr	Unreliable	
99.9%	8.76 hours / yr	Good	
99.95%	4.38 hours / yr	Very Good	
99.99%	52.6 minutes / yr	**Excellent — HA**	
99.999%	5.26 minutes / yr	**Enterprise — HA + DR**	

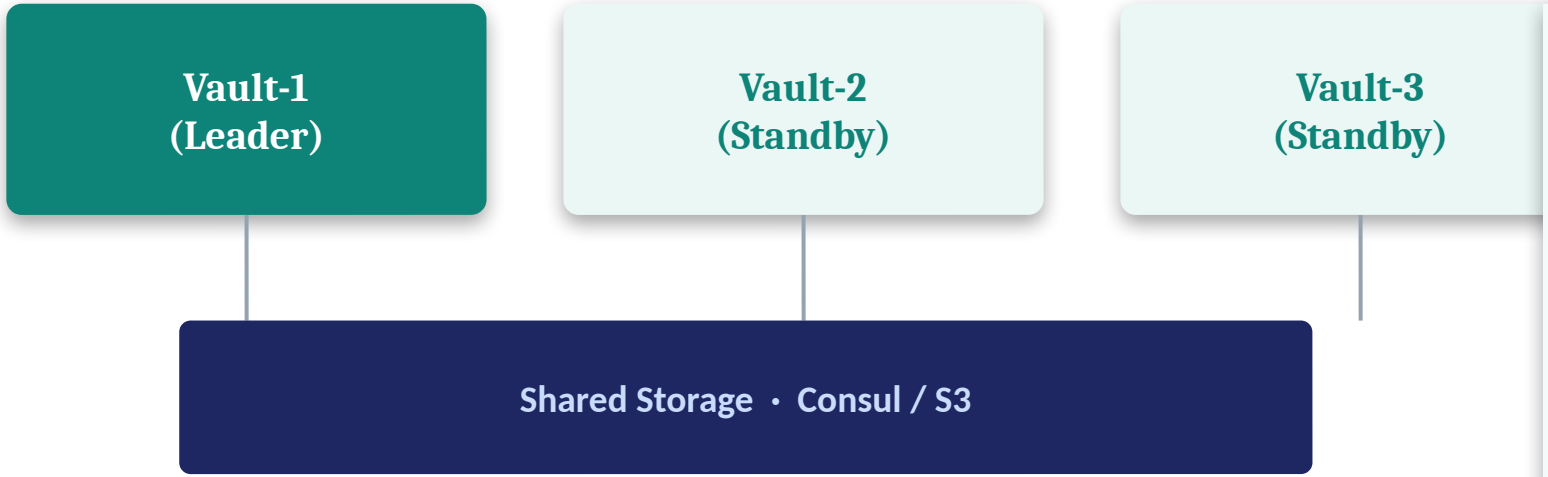
This lab targets 99.99%+ — the HA tier where downtime is measured in minutes per year, not hours.

The Single Node Problem



If Vault dies:

- No access to secrets
- Services can't start
- Complete outage



Only the leader writes data. All nodes can read from shared storage.

If the leader dies:

- Standby promotes to leader
- No secrets lost
- No downtime

Configure & Deploy

```
# /etc/vault/vault.hcl — HA Configuration

storage "consul" {
  address = "consul.devsecops.com:8500"
  path    = "vault/"
}

ha_storage "consul" {
  address = "consul.devsecops.com:8500"
  path    = "vault-ha/"
}
```

1. **Deploy 3 nodes**

   Run vault server on sec-vault-01/02/03 — standbys auto-join via Consul.

2. **Initialize once**

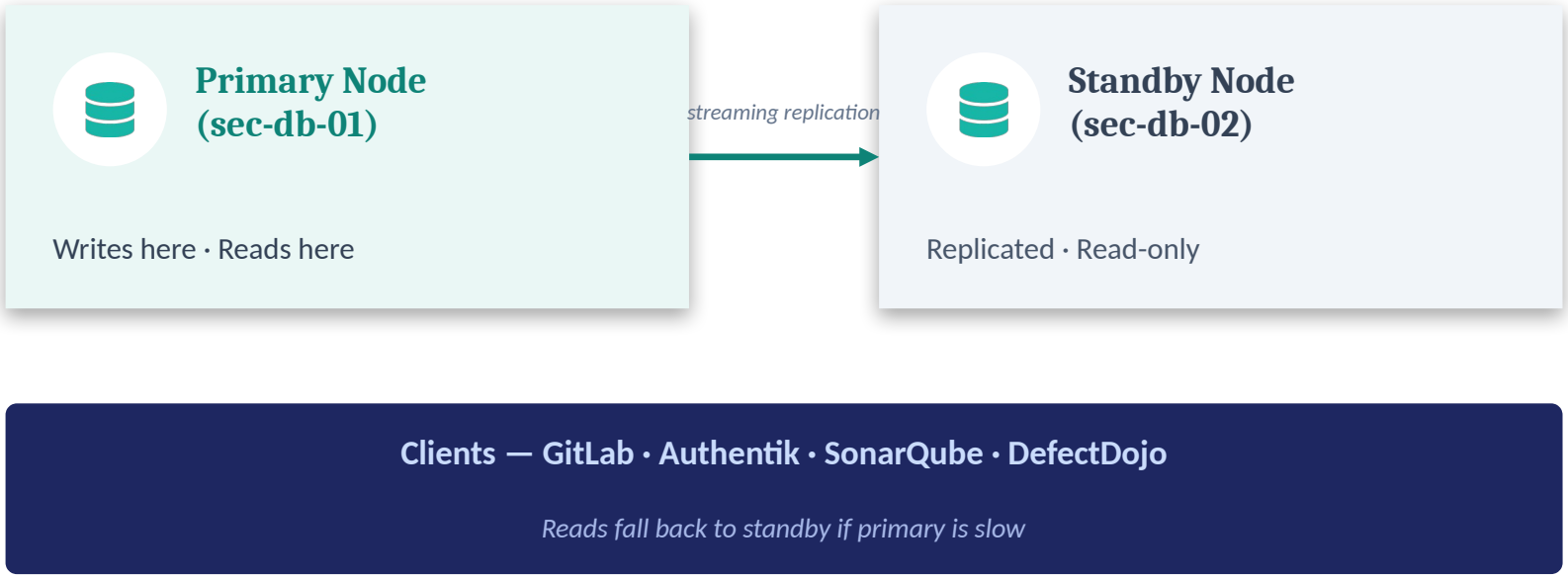
   vault operator init -key-shares=5 -key-threshold=3

3. **Unseal all nodes**

   Run vault operator unseal on each node with the same keys.

4. **Verify HA status**

   vault status → HA Enabled: true, HA Mode: leader / standby.



If primary dies:

- Promote standby to primary
- No data loss
- Automatic via Patroni

Patroni — Automatic Failover

Patroni manages HA PostgreSQL clusters: it watches the primary's health, coordinates leader election through Consul, and promotes a replica automatically when the primary goes down.

```
scope: devsecops-postgresql
name: postgresql-1

restapi:
  listen: 0.0.0.0:8008

consul:
  host: consul.devsecops.com:8500

postgresql:
  data_dir: /var/lib/postgresql/data
  parameters:
    max_connections: 500
```

REST API for health checks

Each node exposes /health for cluster-wide status.

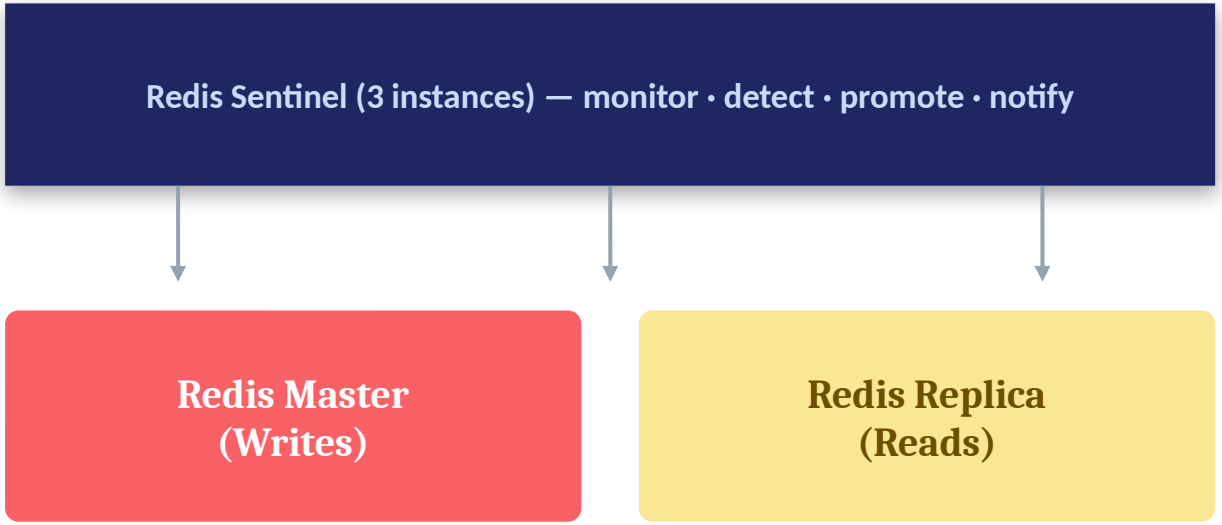
Consul-based leader election

Nodes agree on who holds the lock and is primary.

Automatic replica promotion

pg_ctl_timeout controls how long promotion may take.

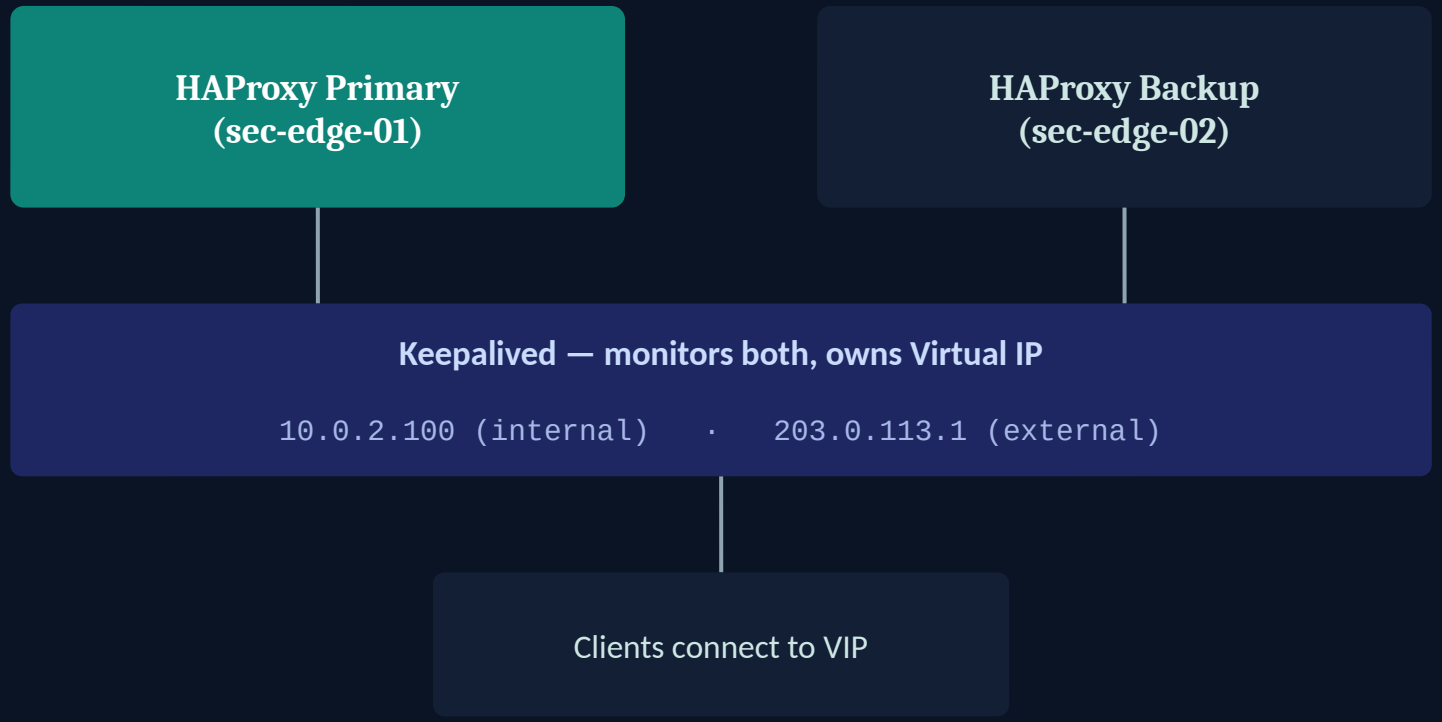
20.4 REDIS HIGH AVAILABILITY



- Sentinel detects in < 1 second
- Promotes replica to master
- Clients auto-reconnect
- Zero downtime (typically)

```
sentinel monitor mymaster 10.0.2.30 6379 2   ·   sentinel failover-timeout mymaster 30000
```

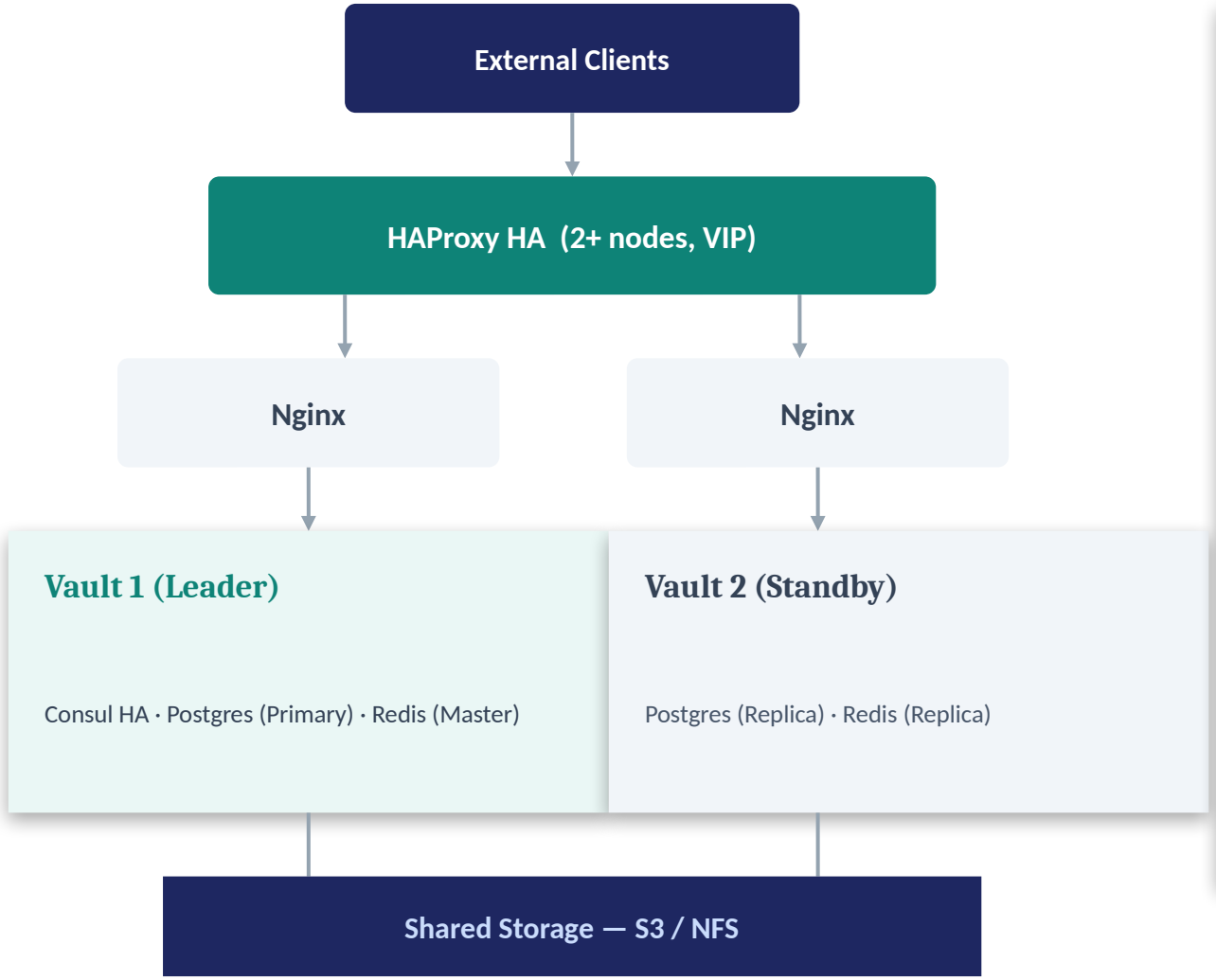
HAProxy + Keepalived



If primary fails:

- Keepalived detects
- VIP moves to backup
- Traffic continues to VIP
- No client config needed

20.6 COMBINED ARCHITECTURE



Result

No single point of failure

Automatic failover

Zero-downtime upgrades

Health Checks & Alerting

Vault Cluster

```
vault operator raft autopilot state
```

Healthy: true · FailureTolerance: 1 · Servers: 3

Replication Lag

```
SELECT NOW() - pg_last_xact_replay_timestamp()
```

Should be < 1 second

Redis Sentinel

```
redis-cli -p 26379 SENTINEL masters
```

Confirms current master + replicas

PROMETHEUS ALERTS

```
VaultClusterUnhealthy
  vault_core_handle_login_request == 0
  for: 5m

PostgresReplicationLagTooHigh
  pg_replication_lag_seconds > 5
  for: 2m

RedisFailoverNeeded
  redis_sentinel_simulate_failure_alarms > 0
  for: 1m
```

20.8 DISASTER RECOVERY TESTING

Vault

```
killall vault
```

Standby promotes → vault status shows new leader

Database

```
systemctl stop postgresql
```

Standby promotes → pg_is_in_recovery() returns false

HAProxy

```
systemctl stop haproxy
```

Keepalived moves VIP → ip addr shows VIP on backup

Redis

```
redis-cli SHUTDOWN
```

Sentinel promotes replica → redis-cli ROLE returns master

20.9 CHAPTER SUMMARY

Component HA Strategy — RPO & RTO

Component	HA Strategy	RPO	RTO
Vault	Clustering (HA mode)	0 (sync)	< 1s
Database	Master-Slave + Patroni	0 (sync)	1–5s
Cache	Sentinel + Replicas	< 1s	< 1s
LB	Keepalived + VIP	N/A	< 1s

RPO = Recovery Point Objective (data loss tolerance) *RTO = Recovery Time Objective (recovery duration)*

KEY TAKEAWAY

Eliminate Single Points of Failure

BEFORE HA

- One node dies = complete outage
- Manual intervention required
- Data loss possible

AFTER HA

- N-1 nodes can fail (quorum-based)
- Automatic failover in seconds
- Zero data loss (sync replication)